technische universität
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AI-Generated Financial Misinformation

Detection, Market Prediction, and Financial Risk Analysis

Data & AI in Economics — TU Dortmund

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Team: Group M
Prathuasha Kandula Bindhu
Roshini Venkata Ramana
Anjali Karna

Motivation & Economic Relevance

Problem

– Financial markets increasingly depend on digital information.

– Generative AI can produce realistic financial misinformation at scale.

Research Question

Can advanced machine learning models reliably detect AI-generated financial misinformation, and can textual features extracted from financial news improve the prediction of stock market outcomes such as returns, trading volume, and volatility?

Economic Relevance

– Fake financial news can shift investor sentiment.

– AI-generated content challenges traditional verification methods.

– Quantifying information impact supports financial-risk assessment.

How misinformation propagates

AI-generated news

↓

Investor sentiment

↓

Market reaction

↓

Financial risk

Data & Methodology

Data

Financial News

- Real financial articles
- Company-linked news data

Synthetic Data

- LLM misinformation – GPT-4.1-mini
- Bullish | Bearish | Neutral rewrite

Market Data

- Yahoo Finance via yfinance
- Returns | Trading volume | Volatility

Methodology

Real financial news

↓

Build synthetic misinformation scenarios

↓

Feature extraction — sentiment, FinBERT, linguistic

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Detection models learn AI-generated patterns

Logistic Regression | Random Forest | Support Vector Machine | MLP Neural Network | FinBERT

↓

Causal analysis on market data

DoWhy | DAG | Backdoor adjustment

Results

AI Misinformation detection performance and causal effect

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.800	0.789	1.000	0.882	0.850
Random Forest	0.825	0.960	0.800	0.873	0.911
Tuned Random Forest	**0.875**	**0.963**	**0.867**	**0.912**	**0.938**
SVM	0.825	0.848	0.933	0.889	0.878
Tuned SVM	0.825	0.848	0.933	0.889	0.878
Neural Network	0.800	0.814	0.950	0.877	0.836
Tuned Neural Network	0.763	0.847	0.833	0.840	0.837
FinBERT	**0.950**	**0.967**	**0.967**	**0.967**	**0.981**

FinBERT is the strongest detector (F1 = 0.97): transformer semantics beat classical ML.

Causal Analysis

Effect of news sentiment on future returns

ATE = −0.000265

small negative association

Refutation : p = 0.98

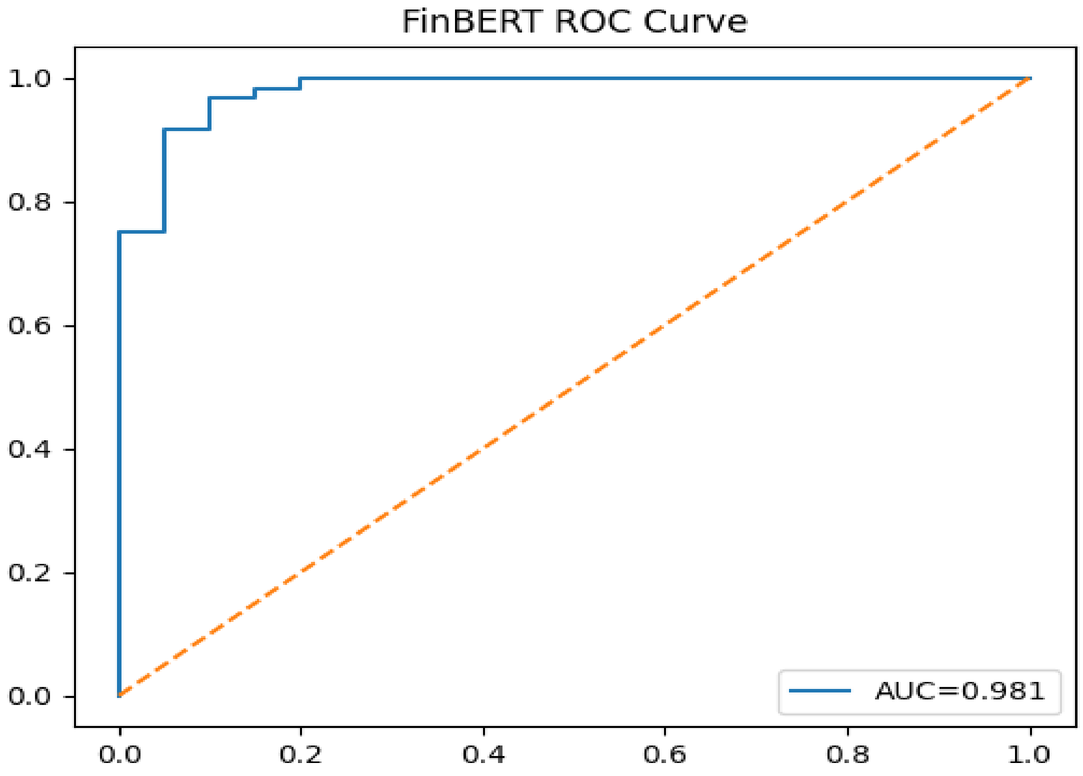
→ ***effect remains stable***

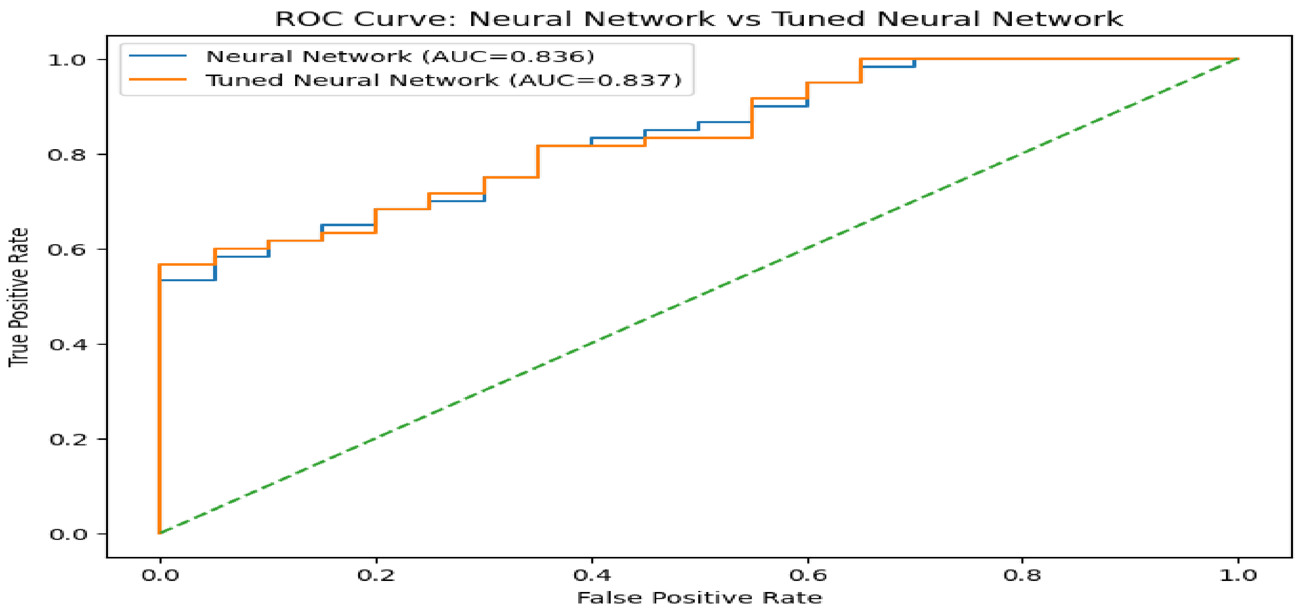
Robustness Evaluation

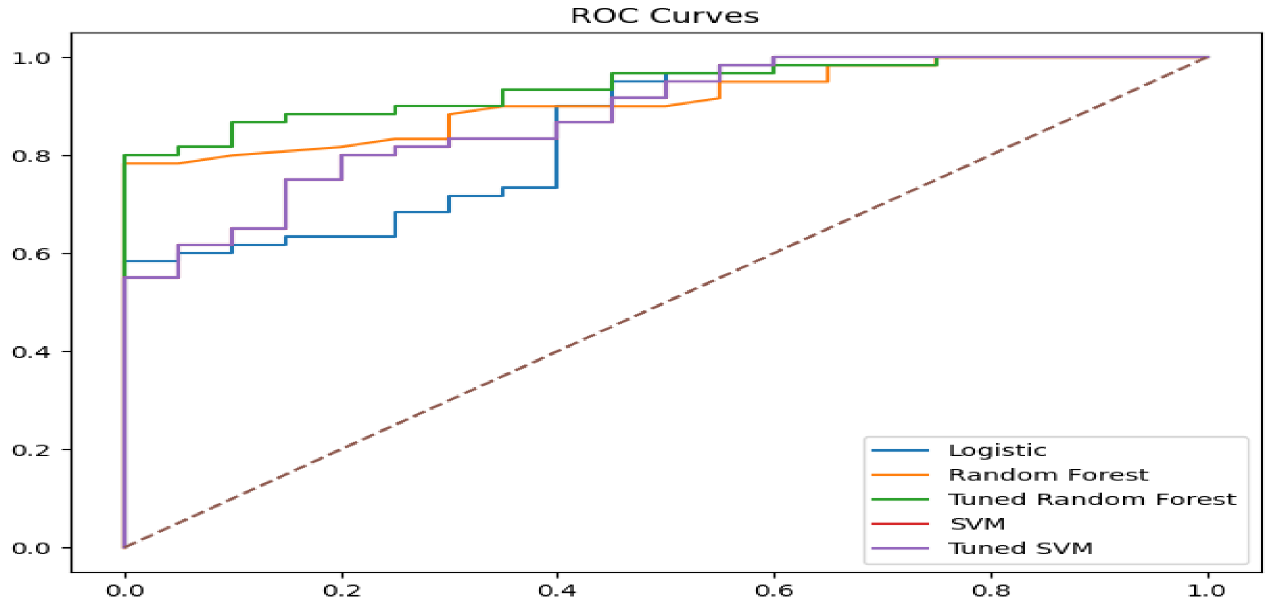
- FinBERT consistently outperformed baseline models.
- Bullish and bearish misinformation were detected reliably.
- Neutral AI rewrites produced most false negatives

ROC Curve Comparison

ROC curve comparison demonstrating progressive improvements from baseline machine learning models to Neural Networks, with FinBERT achieving the best discriminative performance (ROC-AUC = 0.981).

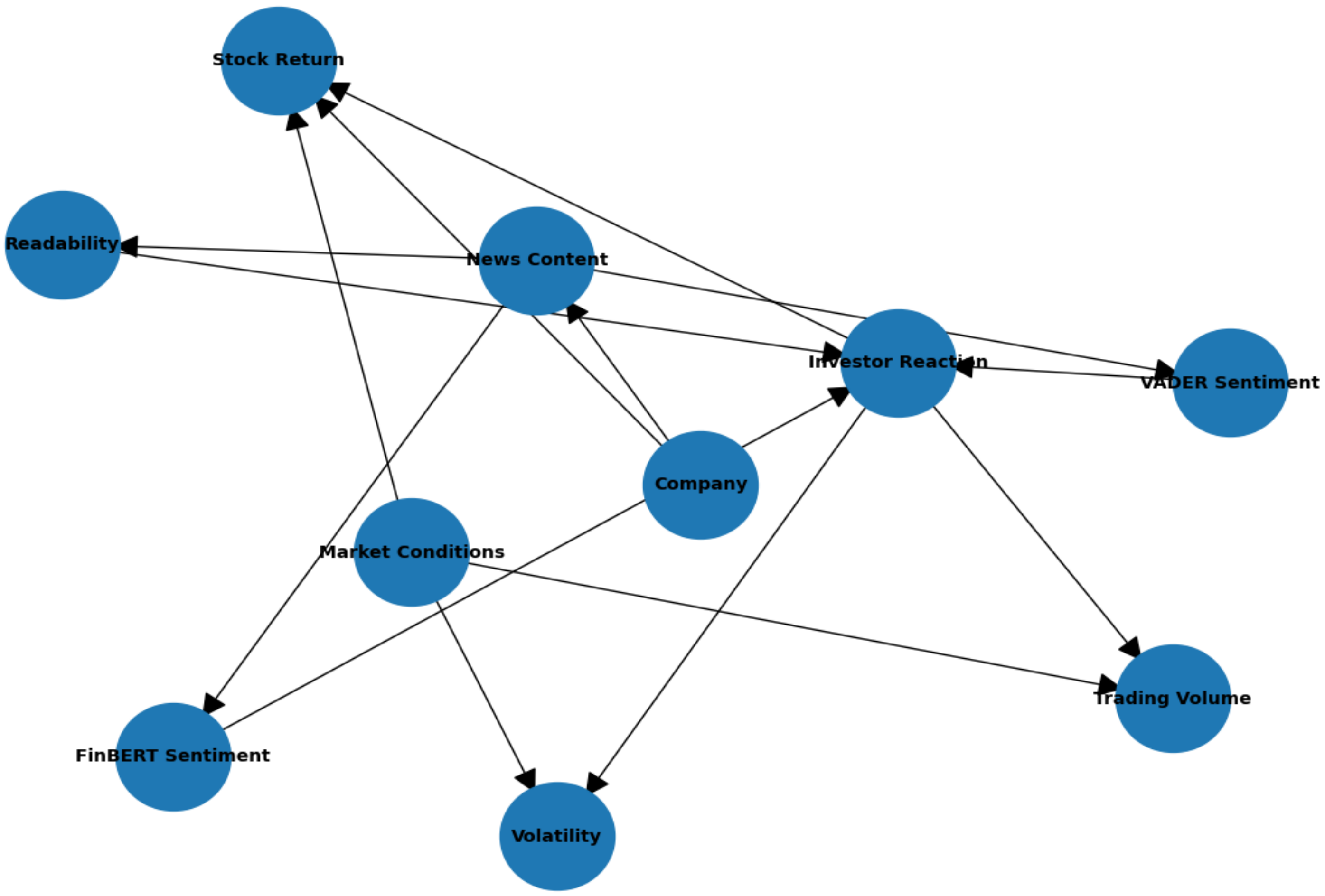






DAG (Causal Graph)

Conceptual DAG for Financial News and Market Outcomes



Limitations & Synthesis

Generative AI

Large Language Models generated realistic bullish, bearish, and neutral financial misinformation. Neutral AI rewrites closely resembled authentic financial news, making detection more challenging.

Machine Learning (Supervised ML)

Traditional machine learning models provided strong baseline performance. FinBERT achieved the highest detection accuracy, demonstrating the advantage of domain-specific transformer models for financial text.

Causal Analysis

Financial news sentiment showed a small negative association with subsequent stock returns after adjusting for observed confounders. Robustness checks supported the stability of the estimated causal effect, although the economic impact remained limited.

Limitations

- Synthetic data cannot fully represent real-world attacks.
- Causal inference depends on the observed confounders.
- Market movements are driven by many external factors.
- Detection may degrade as future LLMs improve.

Conclusion & Key Takeaways

This mission demonstrates that:

- AI-generated financial misinformation can be detected using advanced NLP models.
- Transformer-based models such as FinBERT provide stronger semantic understanding than traditional ML methods.
- Financial text contains useful signals for analysing market behaviour, but misinformation is only one factor affecting returns

References

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